

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("diabetes.csv")
df.head()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI \
0	6	148	72	35	0	33.6
1	1	85	66	29	0	26.6
2	8	183	64	0	0	23.3
3	1	89	66	23	94	28.1
4	0	137	40	35	168	43.1

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

```
df.tail(5)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI \
763	10	101	76	48	180	32.9
764	2	122	70	27	0	36.8
765	5	121	72	23	112	26.2
766	1	126	60	0	0	30.1
767	1	93	70	31	0	30.4

	DiabetesPedigreeFunction	Age	Outcome
763	0.171	63	0
764	0.340	27	0
765	0.245	30	0
766	0.349	47	1
767	0.315	23	0

```
df.shape
```

(768, 9)

df.dtypes

```
Pregnancies      int64
Glucose           int64
BloodPressure     int64
SkinThickness     int64
Insulin           int64
BMI               float64
DiabetesPedigreeFunction float64
Age               int64
Outcome           int64
dtype: object
```

df.info()

<class 'pandas.DataFrame'>

RangeIndex: 768 entries, 0 to 767

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	Pregnancies	768 non-null	int64
1	Glucose	768 non-null	int64
2	BloodPressure	768 non-null	int64
3	SkinThickness	768 non-null	int64
4	Insulin	768 non-null	int64
5	BMI	768 non-null	float64
6	DiabetesPedigreeFunction	768 non-null	float64
7	Age	768 non-null	int64
8	Outcome	768 non-null	int64

dtypes: float64(2), int64(7)

memory usage: 54.1 KB

df.describe()

	Pregnancies	Glucose	BloodPressure	SkinThickness
Insulin \				
count	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458
std	3.369578	31.972618	19.355807	15.952218
min	0.000000	0.000000	0.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000
75%	6.000000	140.250000	80.000000	32.000000

```
127.250000
max      17.000000   199.000000   122.000000   99.000000
846.000000
```

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.992578	0.471876	33.240885	0.348958
std	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.078000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	32.000000	0.372500	29.000000	0.000000
75%	36.600000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000

```
print("Mean")
print(df.mean())
print("\nMedian")
print(df.median())
print("\nVariance")
print(df.var())
print("\nStandard Deviation")
print(df.std())
```

```
Mean
Pregnancies      3.845052
Glucose          120.894531
BloodPressure    69.105469
SkinThickness    20.536458
Insulin          79.799479
BMI              31.992578
DiabetesPedigreeFunction  0.471876
Age              33.240885
Outcome          0.348958
dtype: float64
```

```
Median
Pregnancies      3.0000
Glucose          117.0000
BloodPressure    72.0000
SkinThickness    23.0000
Insulin          30.5000
BMI              32.0000
DiabetesPedigreeFunction  0.3725
Age              29.0000
Outcome          0.0000
dtype: float64
```

```
Variance
Pregnancies      11.354056
Glucose          1022.248314
```

```
BloodPressure      374.647271
SkinThickness      254.473245
Insulin            13281.180078
BMI                62.159984
DiabetesPedigreeFunction  0.109779
Age                138.303046
Outcome            0.227483
dtype: float64
```

```
Standard Deviation
Pregnancies        3.369578
Glucose            31.972618
BloodPressure      19.355807
SkinThickness      15.952218
Insulin            115.244002
BMI                7.884160
DiabetesPedigreeFunction  0.331329
Age                11.760232
Outcome            0.476951
dtype: float64
```

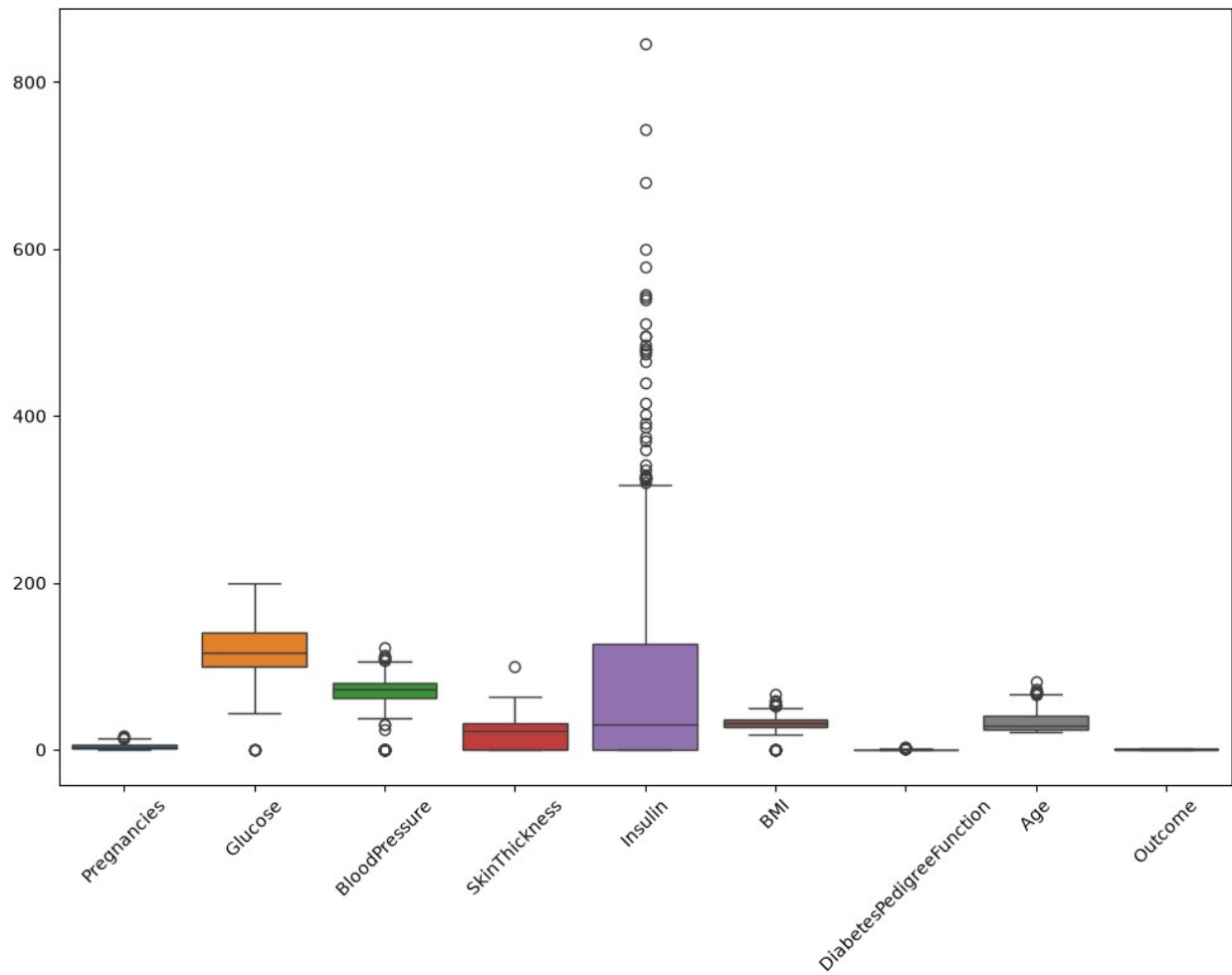
```
df.isnull().sum()
```

```
Pregnancies      0
Glucose           0
BloodPressure     0
SkinThickness     0
Insulin           0
BMI               0
DiabetesPedigreeFunction  0
Age               0
Outcome           0
dtype: int64
```

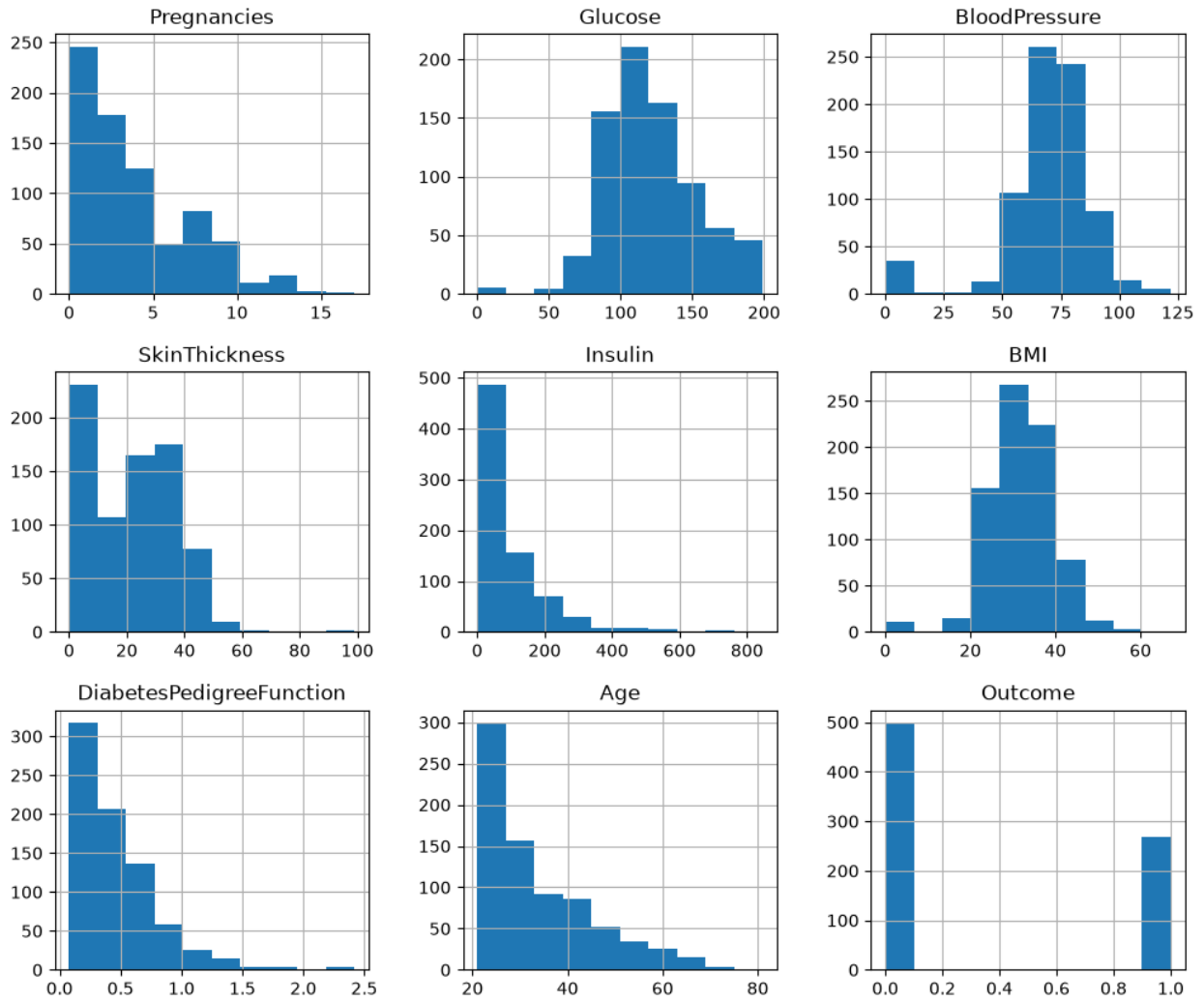
```
columns = ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI']
for col in columns:
    print(col, (df[col]==0).sum())
```

```
Glucose 5
BloodPressure 35
SkinThickness 227
Insulin 374
BMI 11
```

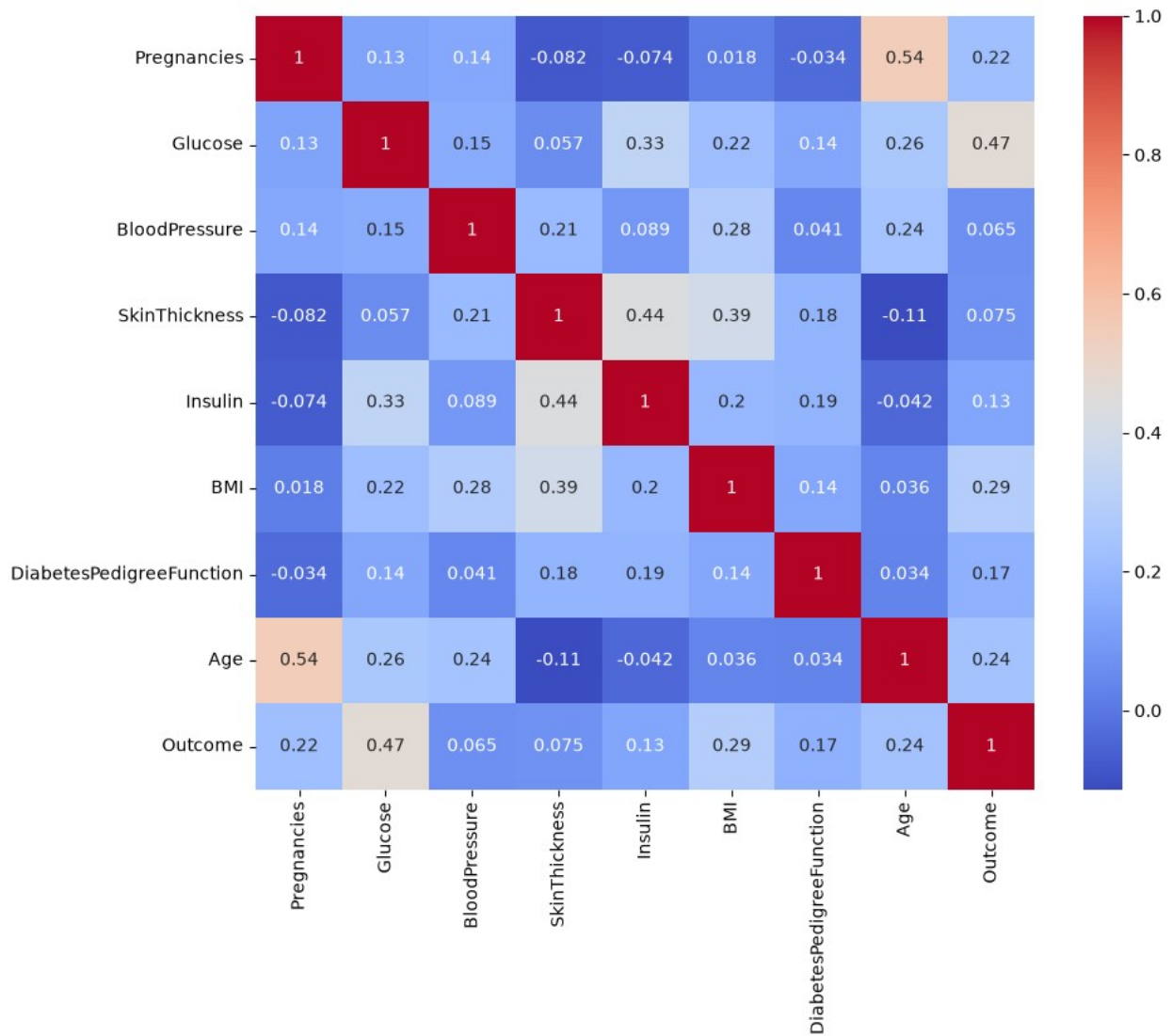
```
plt.figure(figsize=(12,8))
sns.boxplot(data=df)
plt.xticks(rotation=45)
plt.show()
```



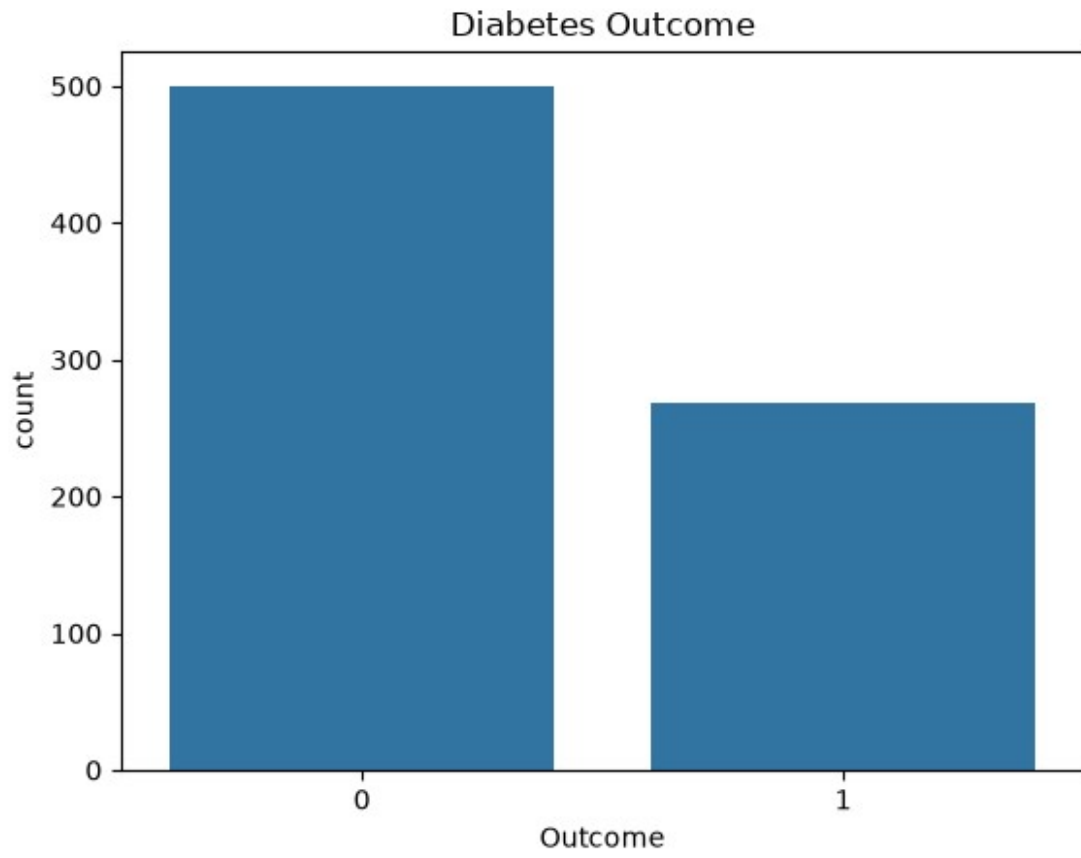
```
df.hist(figsize=(12,10))  
plt.show()
```



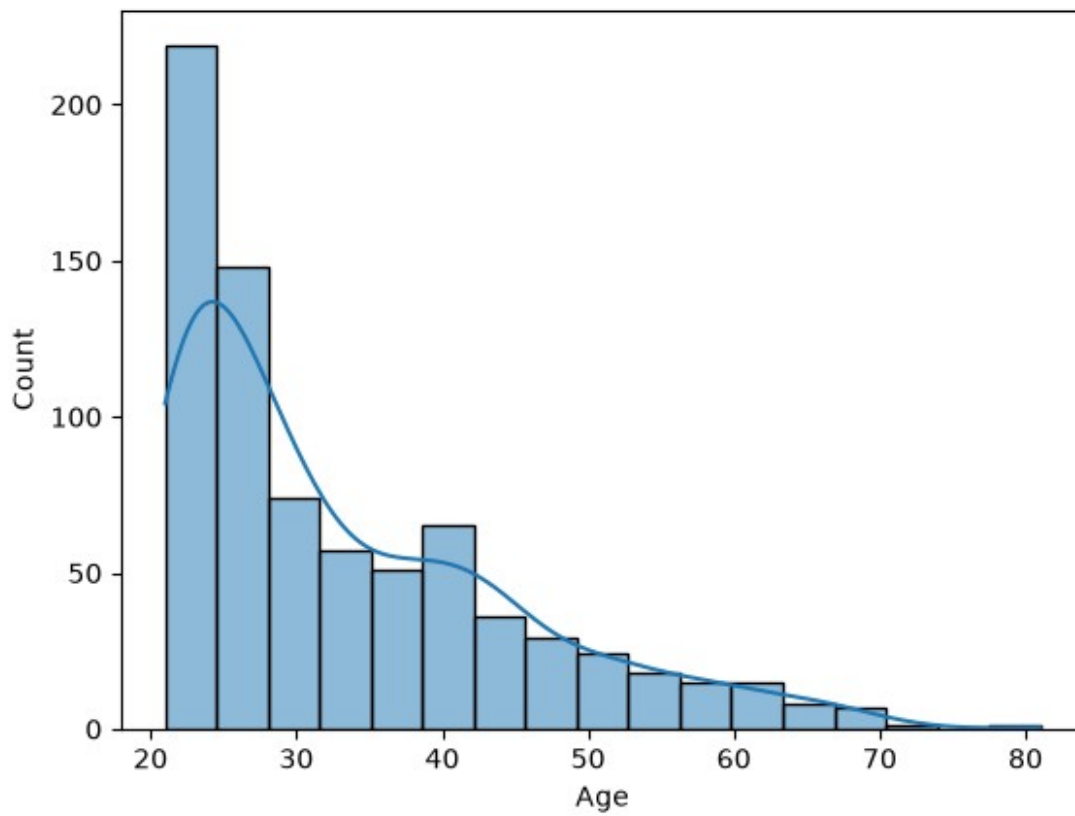
```
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.show()
```



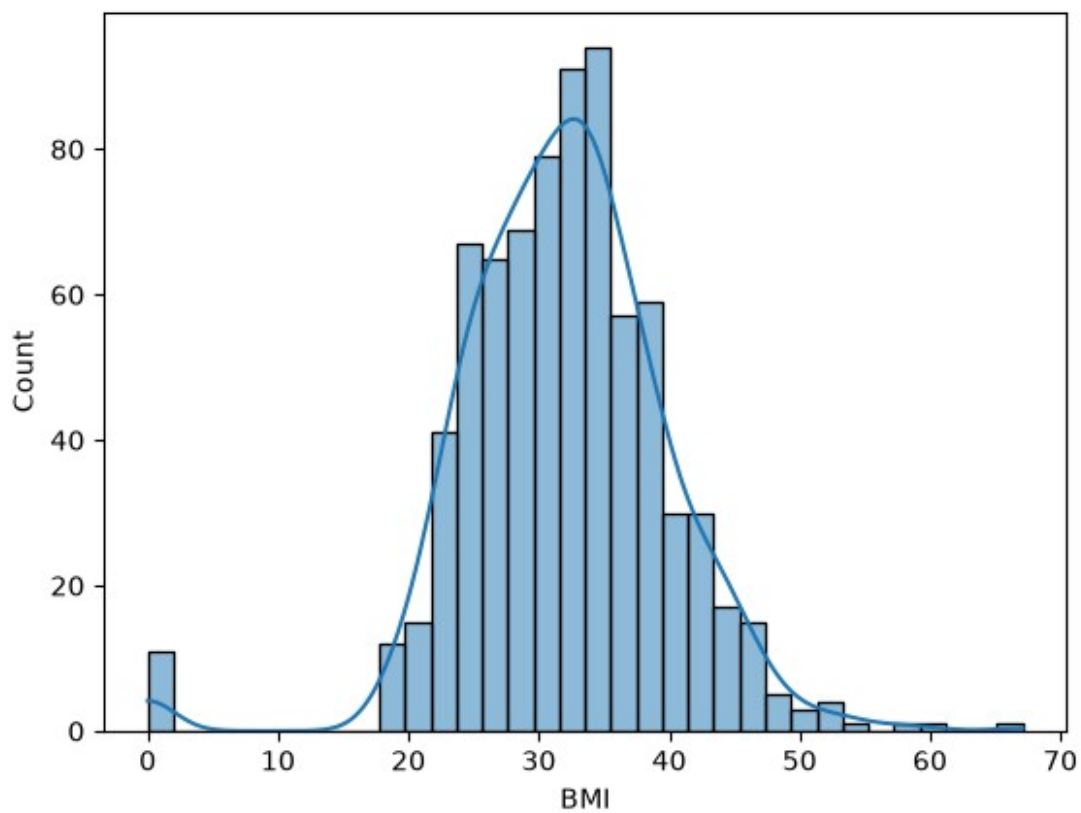
```
sns.countplot(x='Outcome', data=df)
plt.title("Diabetes Outcome")
plt.show()
```



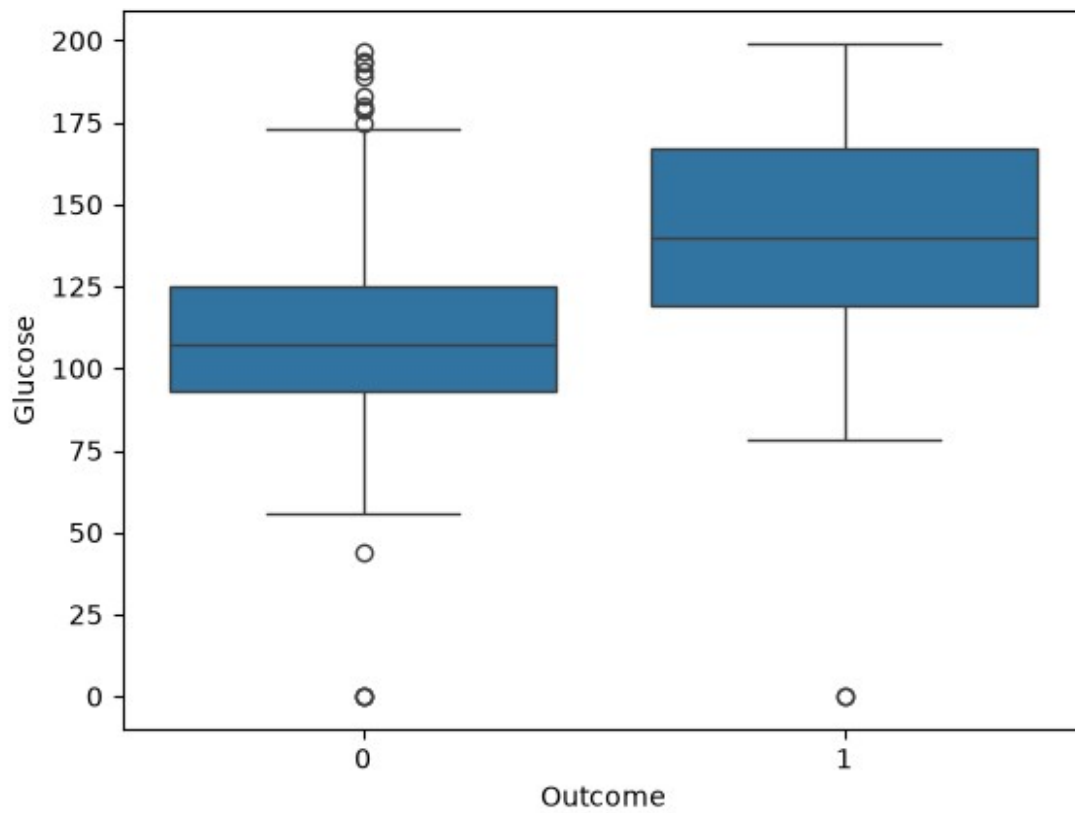
```
sns.histplot(df['Age'], kde=True)  
plt.show()
```



```
sns.histplot(df['BMI'], kde=True)  
plt.show()
```



```
sns.boxplot(x='Outcome', y='Glucose', data=df)  
plt.show()
```



```
sns.pairplot(df, hue='Outcome')  
plt.show()
```

